The Hidden Variable in Embedded VLM Perception: Temporal Calibration Across Models and Platforms for Work-Zone State Estimation

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Paper under double-blind review

***Abstract*—Temporal smoothing constants are a hidden variable that decides comparisons between embedded perception models and between platforms, yet comparisons usually keep each system's published constants. We run a fine-tuned 2B vision-language model (VLM), a zero-shot 4B reasoner (C3E) and a detector+CLIP pipeline on ROADWork work-zone driving on a Jetson AGX Orin and a DRIVE AGX Thor, letting measured latency drop camera frames. First, constants do not transfer between models: inherited from the fine-tuned VLM, they leave C3E at macro-F1 0.290; recalibrated, it reaches 0.515, and the gap holds on unseen drives. Second, the baseline needs the same treatment: recalibrated, it rises from 0.470 to 0.551, no VLM-based system beats it, and a counted hidden-Markov filter on its detector score alone is not significantly worse. Third, constants do not reliably transfer between platforms: on the Thor, carried-over constants cost the hand-tuned pipeline and C3E 0.03–0.04 macro-F1. The recalibrated pipeline and the per-second counted filters keep their macro-F1 there, but the pipeline gains 15 false alarms per hour while the detector-only filter loses 4. The first two findings hold on drive-disjoint folds and at matched false-alarm rates, and live runs reproduce the replay.**

I. Introduction

Work zones concentrate difficult driving-perception conditions: their appearance is regional and unstandardized, geometry changes daily, evidence such as cones, barrels, signs, and workers is small or occluded, and both misses and nuisance alerts are costly. They accounted for roughly 101,000 U.S. crashes and 899 fatalities in 2023 [1], yet production ADAS rarely model work zones explicitly and temporary speed limits are often absent from digital maps.

The strongest onboard approach combines a YOLO-style work-zone detector, CLIP verification [2], exponential smoothing, and hysteresis to estimate four traversal states [3], [4]. It is fast and effective after a zone is established, but reported IoU falls from 0.56/0.47 on OUTSIDE/INSIDE to 0.11/0.09 on APPROACHING/EXITING [3]. These boundaries are precisely where an advisory is most useful: an alert after the vehicle reaches the barrels has little anticipatory value. The benchmark is hard at these boundaries: under the macro-F1 we use, the published per-state IoUs correspond to 0.43 (the paper reports 0.600 under its own aggregation), and our port of the system scores 0.470. Every system we test separates work-zone from outside frames about equally well (zone F1 0.84–0.86), so systems differ in *when* they switch states.

Whether VLMs should replace detector pipelines is usually judged by comparing systems as published, each with its authors' own smoothing and hysteresis constants. Fine-tuned VLMs describe work-zone scenes well offline [5], and weight-only quantization plus embedded runtimes bring small VLMs to automotive hardware [6], [7]. We build the strongest embedded VLM estimator we can, run it beside the detector pipeline on the same videos and protocol on two platforms, a Jetson AGX Orin and the next-generation DRIVE AGX Thor, and ask what such a comparison actually measures.

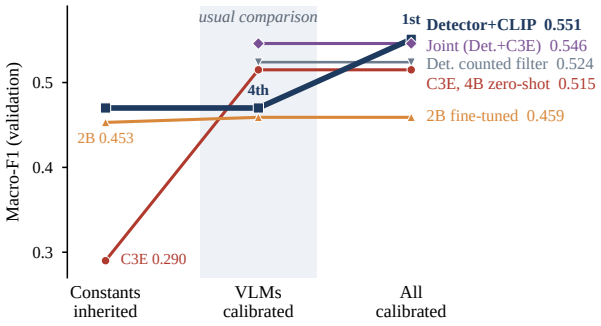


Fig. 1. The hidden variable: validation macro-F1 under three comparison regimes. Left: inherited constants make the zero-shot C3E look useless. Middle (the usual comparison): calibrated VLM systems beat the published detector pipeline. Right: calibrating the baseline by the same procedure makes it the best system. Grey: the detector-only counted filter, which carries most of the joint estimator's lead.

The answer is: mostly the temporal constants (Fig. 1). Against the hand-tuned detector pipeline our VLM-based systems look competitive or better, but calibrating the baseline by the same procedure reverses the ranking. A team choosing a perception stack from the usual comparison would adopt a costlier VLM stack for a gain that disappears once the baseline is calibrated. The VLM-based systems' edge shows on signled onsets: the joint estimator and C3E warned $\geq$2 s ahead on all ten in validation, the recalibrated pipeline on eight; across all onsets at matched nuisance, the detector-only counted filter warns early most often. Running C3E beside the detector adds only 5.5 W. A per-second counted filter on the detector score alone is not significantly worse than the best system in macro-F1, keeps it across platforms and raises the fewest false alarms on the Thor.

The same variable decides comparisons between platforms. On the faster Thor, constants counted in cycles span less time and can lose macro-F1; the recalibrated pipeline and per-second counted filters keep it, but their false alarms change

in ways a macro-F1 search does not see.

Our contributions are:

- **Evaluation:** a latency-honest replay protocol with per-model and per-platform recalibration, checked against live runs;
- **Findings:** temporal constants transfer neither between models nor between platforms; calibrating the baseline removes the VLM-based lead, also on unseen drives and at matched false alarms; after a platform change, false alarms must be rechecked even where macro-F1 transfers;
- **Systems:** an evidence-first cascade for a quantized 2B VLM, an INT4 Cosmos3-Edge reasoner on both platforms, and a per-second counted hidden-Markov filter that keeps its macro-F1 across platforms, with an analysis of the VLMs' ego-relevance gap.

II. Related Work

a) Work-zone perception.: Early systems used environmental models with sensor fusion [8] or frame classification [9]. ROADWork [5] provides 4,375 videos and 9.6K annotated images from 18 U.S. cities and shows large gains from in-domain fine-tuning (+32.2 AP, +36.7 SPICE). The closest system [3] deploys a ROADWork-trained YOLO detector with CLIP verification, EMA smoothing and hysteresis, and defines the traversal-state evaluation we adopt.

b) VLMs and world models for driving.: VLMs support planning [10], [11], description [12] and reasoning [13], and Cosmos world models plan at datacenter or automotive-SoC scale [14], [15]; we estimate one safety-relevant state under an embedded latency budget. Unlike detection [16] and LLM [17] cascades, which chain models of increasing cost and escalate on confidence, our stages share one model and escalate on evidence semantics (transcribed text versus a binary answer).

c) Temporal filtering and fair comparison.: That baselines deserve the proposed method's tuning budget is known [18]; we measure how much it changes conclusions for embedded temporal estimators. Our joint estimator is a discrete hidden Markov model [19] run as a Bayes forward filter [20]; we claim no novelty for it.

III. System

A. Two embedded deployments

We deploy *two* models on the same Jetson AGX Orin 64 GB, which shares its SoC with the automotive DRIVE AGX Orin, and rerun them on a DRIVE AGX Thor (Section IX). They represent two common ways to adapt a VLM: task fine-tuning, and quantization with per-model calibration alone. Both run the same prompt channels, cascade structure, protocol and splits; only the model, its input size and its calibrated constants differ.

Fine-tuned 2B. A 2B-parameter VLM (Qwen3-VL architecture [21], initialized from Cosmos-Reason2-2B [22]) serves for perception only. It is fine-tuned through a curriculum derived from ROADWork [5], whose imagery spans 18 U.S. cities under one signing standard (MUTCD [23]). The curriculum covers work-zone VQA ($\sim$46K pairs), ego-state supervision, sign reading, a general-driving retention mix, and $\sim$10K work-zone-free BDD100K hard negatives. The last stage cut the existence gate's "yes" rate on clean BDD100K scenes from 95% to 0%. On ROADWork OUTSIDE seconds, which include visible but irrelevant work, the gate is less selective (Section X).

Decoding, not vision encoding, dominates the 2B's cost: 10.4 ms per token, against $\sim$20 ms per image at our 480px. The language backbone is quantized to INT4 (W4A16 AWQ) and the vision tower kept in FP16, both as on-device TensorRT engines [7].

Quantized 4B world model. Cosmos3-Edge (C3E) [14] pairs an autoregressive reasoner with a trajectory-generating diffusion transformer; we run only the reasoner, and "world model" below refers to it. It is quantized to W4A16 INT4 with an FP16 visual tower on the same Orin. Its frozen 26×46 patch grid fixes inputs at 736×416. It has seen no work-zone training data. Its engines required six patches to the embedded runtime, which we release (Section XII).

B. Prompt channels and their failure modes

Each of our four fixed-prompt channels exists because of a failure mode we observed. All use fixed token budgets, and the first generated token is an answer-start token.

a) Affirmation bias.: Binary prompts cannot be specialized by rephrasing. The GATE asks a neutral existence question, but path-relevance variants all answered "yes" on OUTSIDE probes and distance prompts "close" regardless of scene. Binary outputs are therefore weak evidence requiring a temporal vote, never sole triggers. A three-token answer costs $\sim$85 ms and a description $\sim$500 ms, so evidence quality, not only confidence, gates smoothing.

b) Faithful transcription.: The SIGN channel returns verbatim text such as "SHOULDER WORK" and correctly reports absence, while a binary rephrasing misses signs that transcription catches; we match transcripts against a work-zone lexicon.

c) Structured corroboration.: DESC often verbalizes disqualifying context ("barriers on right *sidewalk*", "work vehicle *parked*"), so corroboration requires a work-zone object without such a qualifier. Prompts, budgets and lexicons are part of our release (Section XII).

C. Evidence-first cascade

The cascade (Fig. 2) allocates compute by state. In OUTSIDE, GATE and SIGN run every cycle, which we call the patrol cycle (Section V-C), and either can invoke DESC; fast entry on a transcribed "ROAD WORK AHEAD" avoids a voting delay of up to N cycles ($\sim$1.3 s). In active states, EGO updates a Bayesian filter whose argmax is restricted to the active states, so one "outside" reply right after a sign-triggered entry cannot erase it. Only sustained gate clearance returns to OUTSIDE; partial fading triggers EXITING, which EGO never produced. The 2B's constants ($N/K_{\text{enter}}/K_{\text{exit}}/K_{\text{fade}}$ = 5/3/4/2) were fixed on calibration data; C3E's (3/3/2/1) come from Section VI. With EGO disabled, as selected for C3E,

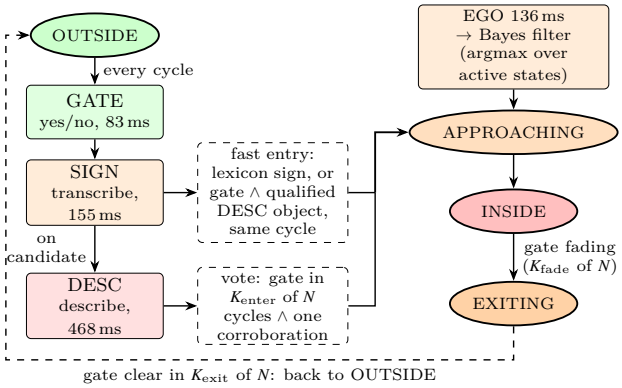


Fig. 2. Evidence-first cascade (2B latencies, p50). Entry needs a sustained, corroborated vote or high-confidence same-cycle evidence. The constants N, K_{enter}, K_{exit}, K_{fade} are calibrated per model: the structure transfers, the numbers do not.

APPROACHING becomes INSIDE on the transition prior, 13 cycles after entry.

D. Joint estimator over both evidence streams

Grafting the detector and the world model onto each other's temporal machinery does not work. The ported detector counts cycles (150-cycle maximum approach, 20 minimum frames outside) tuned for its native loop. The cascade votes over a window tuned for the 2B's ∼260 ms cycle, while C3E answers only every ∼1.5 s. Either spine imports constants that are wrong for the combination.

The joint estimator therefore inherits no constant. Its observation is a lookup cell of discretized evidence: the detector's frame score (six buckets), the world model's gate, sign-keyword and corroboration answers ordered by strength (six levels), and the age of that answer (three buckets); only 20 of the 108 cells hold 97% of the observations. A discrete hidden Markov model over the four states runs on these cells as a forward filter [19], with maximum-likelihood tables and no EMA, hysteresis threshold or cycle counter. We call it a counted filter because its emission and transition tables are counted from calibration data, with transitions as per-second rates; run on the detector score alone, it is the detector-only counted filter. The emission is counted once per detector cycle with Laplace smoothing, jointly rather than factorized, because the interaction is the signal: the world model's APPROACHING is reliable when the detector corroborates it and mostly a false alarm when it does not (73% of those frames are OUTSIDE). The transition is a per-second rate matrix, counted as ground-truth transitions over dwell time and scaled by each step's dt, so it does not depend on the loop rate. The log-likelihood is scaled by dt/τ ($\tau = 0.5$ s) because a world-model answer persists across ∼30 detector frames and would otherwise be counted that many times. Bucket edges and τ were set on calibration data (Section VII).

The detector runs at its native rate and never waits for the world model; the world model is re-polled as soon as it answers, and its last answer is held with its age. We record raw evidence with measured clocks once per video and replay the filter on it, so ages and frame drops in every result are measured ones; Section VII confirms the replay against live concurrent execution.

IV. Evaluation Protocol

a) Benchmark.: We evaluate on the baseline's own benchmark, with every design choice made on a calibration split. The ROADWork traversal benchmark [3] has 520 videos (4.31 h of driving) from 25 drives, with 484K interval-labeled frames. Videos are 1920×1080; all run at 30 fps except seven at 120 fps. A state-stratified 60/40 split separates calibration from validation, on which all results are reported. Calibration has 312 videos (2.59 h). Validation has 208 videos (1.73 h, 192K frames), 13 of them pure-negative. All 25 drives occur in both partitions, so Section VIII repeats every replay-fitted comparison with drive-disjoint folds. ROADWork samples its annotated images from the same drives as its videos, so the 2B's fine-tuning images, like the baseline detector's training images, may come from benchmark drives; C3E is zero-shot. Every calibration effect we report compares a system with itself, perception fixed, and is not inflated by this overlap.

b) Latency-honest camera simulation.: Slower systems pay for their latency: the replay drops the frames a live camera would miss. After each measured inference cycle of duration Δ on the target hardware, the replay advances by Δ, so slower systems see sparser frames and their latency enters the accuracy result. Predictions are held between cycles and scored per frame.

c) Metrics.: Macro-F1 over the four states is the headline metric; false alarms per hour measure nuisance, and Lead measures anticipation. We follow the baseline's suite [3]. Frame accuracy and macro-F1 (the mean of per-state frame F1, $F_1 = 2\,\text{IoU}/(1+\text{IoU})$) score every frame; zone F1 is the frame F1 of any active state vs. OUTSIDE. We also report per-state IoU and event precision/recall, where an interval is correct if it overlaps one of the same state. Entry offsets are the Entry MAE over all states and the median INSIDE entry offset (IN entry med.). False alarms per hour count predicted active intervals that overlap no annotated one. Lead is the share of INSIDE onsets preceded by ≥2 s of active state. F1@20 and Lead@20 repeat macro-F1 and Lead after one causal persistence filter, which shows an active interval only after it lasts d seconds, with d the per-system operating point at which false alarms reach 20 per hour. Accuracy, precision, recall and Lead are in percent. The replay runs each video at its native frame rate. Per-hour rates count frames at 30 fps (validation time overstated by 3%, for all systems alike). Unless stated otherwise, all intervals are drive-level 95% paired cluster-bootstrap intervals (10,000 replications that resample drives), and "significant" means that the interval excludes zero; in tables, † marks such an interval; a difference whose interval includes zero is a tie in the sense of not significantly different.

TABLE I
ORIN AND THOR RESULTS ON VALIDATION (208 VIDEOS), ONE PROTOCOL.

Metric	Det.+CLIP hand	Det.+CLIP recal.	Det. counted	2B	C3E	Joint
Frame acc.	62.9%	64.8%	**68.2%**	62.0%	65.5%	66.1%
Macro F1	0.470	**0.551**	0.524	0.453	0.515	0.546
Zone F1	0.847	**0.860**	0.843	0.839	0.853	0.855
IoU OUT	0.617	0.627	**0.652**	0.581	0.615	0.641
IoU APPR	0.124	**0.296**	0.219	0.152	0.278	0.246
IoU IN	0.544	0.552	0.573	0.532	0.574	**0.578**
IoU EXIT	0.106	0.153	0.124	0.062	0.071	**0.160**
IN ev. recall	95.7	77.8	88.6	**97.3**	87.0	85.9
IN ev. prec.	79.9	**92.5**	75.9	56.8	71.5	66.8
False alarms/h	30.9	**20.2**	22.5	56.2	64.6	25.9
Entry MAE (s)	2.52	2.31	2.44	2.52	**2.15**	2.22
IN entry med. (s)	3.47	**2.10**	2.73	2.67	2.50	2.27
F1@20	0.410	0.529	0.520	0.409	0.487	**0.537**
Lead@20	28.7	60.6	**71.9**	42.5	45.6	61.9
Persistence d	5	1.5	0.25	3	3.25	0.5
Thor macro F1	0.439	0.541	0.522	0.446	0.473	**0.549**
Thor FA/h	52.3	35.4	**21.4**	77.6	105.7	35.4

hand: hand-tuned [3]; recal.: recalibrated; Det. counted: detector-score counted filter; 2B: greedy, no post-filter; Joint: detector+C3E counted filter; Thor rows live except recal. and Det. counted (replays, Section IX).

TABLE II
PAIRED COMPARISONS ON VALIDATION (208 VIDEOS) UNLESS MARKED: MACRO-F1 OF A AND B, AND DIFFERENCES B−A WITH DRIVE-LEVEL 95% INTERVALS.

A → B	F1	ΔF1	ΔFA/h
Models			
C3E: 2B's→own	0.290→0.515		
held-out drives[a]	0.268→0.485	$+0.217^{\dagger}$ [+0.190, +0.244]	$+12.9^{\dagger}$
2B base→fine-tuned[b]	0.377→0.493	$+0.117^{\dagger}$ [+0.039, +0.185]	
2B original→recal.	0.453→0.459	+0.006	
held-out drives[a]	0.464→0.490	$+0.025^{\dagger}$ [+0.001, +0.049]	+6.3
Baseline			
Det.+CLIP hand→recal.	0.470→0.551	$+0.081^{\dagger}$ [+0.043, +0.111]	
VLM search budget	0.470→0.522	$+0.052^{\dagger}$ [+0.038, +0.070]	
unseen drives[c]	0.481→0.535	$+0.054^{\dagger}$ [+0.026, +0.078]	
Score hand→recal.[c]	0.434→0.523	$+0.089^{\dagger}$ [+0.058, +0.114]	
Recal.→C3E	0.551→0.515	−0.036 [−0.080, +0.005]	
Hand→Joint	0.470→0.546	$+0.076^{\dagger}$ [+0.038, +0.104]	
C3E→Joint	0.515→0.546	$+0.031^{\dagger}$ [+0.006, +0.058]	
Recal.→Joint	0.551→0.546	−0.005 [−0.046, +0.033]	
unseen drives[c]	0.535→0.537	+0.002	
Recal.→Det. cnt.	0.551→0.524	−0.026 [−0.062, +0.009]	+2.2 [−8.1, +11.1]
unseen drives[c]	0.535→0.524	−0.010 [−0.036, +0.017]	
Score recal.→Det. cnt.			$-11.2^{\dagger}$ [−24.6, −2.1]

A → B	F1	ΔF1	ΔFA/h
2B cascade ablations			
Binary SIGN	0.464→0.407	$-0.057^{\dagger}$ [−0.092, −0.026]	60.1→305.8
No qualifier filter	0.464→0.447	$-0.017^{\dagger}$ [−0.029, −0.009]	
Unrestricted filter	0.464→0.445	$-0.019^{\dagger}$ [−0.043, −0.001]	
Replay → live			
Joint, Orin	0.546→0.548	+0.002 [−0.011, +0.020]	+0.6 [−4.7, +5.6]
Det.+CLIP, hand, Orin	0.473→0.470		
Det.+CLIP, hand, Thor	0.442→0.439		
Orin → Thor, Orin constants			
Det.+CLIP, hand	0.470→0.439	$-0.031^{\dagger}$ [−0.037, −0.023]	
2B	0.453→0.446	−0.007 [−0.021, +0.008]	
C3E	0.515→0.473	$-0.042^{\dagger}$ [−0.055, −0.026]	
Det.+CLIP, recal.[r]	0.551→0.541	−0.010 [−0.029, +0.012]	$+15.2^{\dagger}$ [+8.6, +21.7]
Joint	0.546→0.549	+0.003 [−0.018, +0.028]	
Det. counted[r]	0.524→0.522	−0.002 [−0.005, 0.000]	$-3.9^{\dagger}$ [−7.1, −1.4]
Thor hand→recal.[r]	0.442→0.541	$+0.099^{\dagger}$ [+0.072, +0.119]	−15.2
Thor recal.→Det. cnt.[r]	0.541→0.522	−0.019 [−0.045, +0.007]	$-14.1^{\dagger}$ [−24.1, −3.6]

[a] Drive folds within the 312 calibration videos. [b] Each recalibrated on its own evidence; held-out drives of the search videos. [c] Drive-disjoint 5-fold replay, all 520 videos. [r] Replay of the detector+CLIP recordings of both devices.

V. THE VLM CASCADE AGAINST THE PUBLISHED BASELINE

A. Comparison with the deployed detector system

Against the published pipeline, the fine-tuned 2B is a competitive detector-free result, not a replacement. The greedy 2B ties it on accuracy and macro-F1 (-0.018, $[-0.041, +0.003]$) and matches its INSIDE recall (Table I). It trails on precision and false alarms, largely the ego-relevance errors of Section X; a causal INSIDE debounce recovers much of the precision (Section V-B). Its APPROACHING IoU is higher, as text and distant evidence appear before detector counts stabilize, a gap the recalibrated detector closes (Section VI). Macro-F1 weights the four states equally and is strict at their boundaries: EXITING covers 5% of validation frames in segments of about 2 s (median), so a boundary off by 1 s can halve its IoU.

B. Ablations and design validation

Only three single-component toggles of the sampled cascade (macro-F1 0.464) change macro-F1 significantly: binary SIGN (0.407), removing the qualifier filter (0.447) and the unrestricted filter (0.445), none by more than 0.06. The sign channel must transcribe: an equal-cost binary question cuts accuracy from 62.5% to 52.5% and raises false alarms from

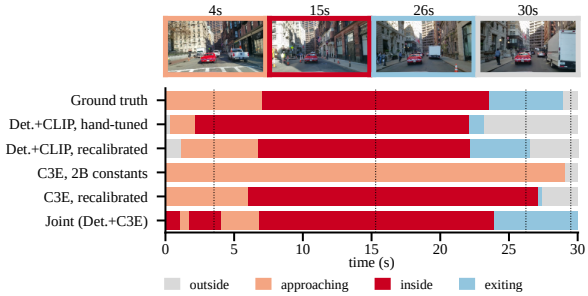


Fig. 3. Ground truth and five systems on one validation video, chosen to illustrate the effect; thumbnails mark the middle of each annotated state. With the 2B's constants, C3E never leaves APPROACHING; recalibrated, it tracks the zone. The hand-tuned detector pipeline enters INSIDE 5 s early and cuts EXITING short; recalibrated, it follows all four states.

60.1 to 305.8 per hour. Greedy decoding gives 0.453 instead of 0.464 but raises INSIDE precision from 54.1% to 56.8% and lowers false alarms, so we use it as the reproducible default; a causal 1 s INSIDE debounce lifts it to 0.478 and 76.0% precision. Fast entry buys anticipation, 10 more points of Lead, for 26 more alarms per hour. The base checkpoint collapses to 0.289 under the 2B's thresholds (Section VI).

C. Operational cost

The cascade keeps pace with a live camera: at p95 latency, patrol and active cycles run at about 4 Hz (259 and 239 ms). This is a pacing guarantee, not an alert deadline: at 30 m/s, the vehicle travels 7.8 m during one p95 patrol cycle. The power cost is modest: above idle, the 2B cascade draws 28.6 W against 22.4 W for the detector, and running C3E beside the detector adds 5.5 W (39.0 vs. 33.5 W median).

VI. Calibrating Every System

Calibrating every system by the same replay procedure reverses the ranking of Section V, where each VLM ran the 2B's thresholds and the detector pipeline its published constants (Fig. 1).

a) Temporal hyper-parameters do not transfer.: With the 2B's constants, the zero-shot C3E almost never reaches INSIDE (IoU 0.007, recall 3.8%) and looks useless. Recalibrating the same four integers and two switches lifts its INSIDE IoU to 0.574 and its macro-F1 from 0.290 to 0.515, with model, prompts and videos unchanged (Fig. 3 and Table II). The search covers $N \in \{3, 5, 7, 9\}$, $K_* \in 1..N$ and the EGO/SIGN switches (4,896 configurations on the 100 calibration videos with recorded VLM evidence; extending it to $N \in \{1, 2, 4\}$ leaves the optimum unchanged). The optimum (N=3, K=3/2/1) is shorter and less hysteretic than the 2B's and disables EGO (Section X). Because each channel is recorded once and the state machine replayed, a sweep costs CPU-seconds instead of GPU-hours.

The same search moves the fine-tuned 2B by only +0.006: it rescues a model whose constants were wrong rather than rewarding flexibility.

b) The baseline too.: The detector pipeline deserves the same treatment. We record every component on every cycle of all 520 videos (detector counts before and after CLIP verification, global CLIP and colour-context scores) and replay the complete pipeline; its hand-tuned configuration reproduces the live run (Table II). Searching its thresholds, EMA rate and duration limits on all 312 calibration videos (three grids of 4,800 configurations) raises validation macro-F1 significantly, by 0.081. With the highest INSIDE precision and the fewest false alarms (Table I), it becomes the strongest system we measure. The same search lifts the detector score alone, without CLIP, by a similar margin (0.435 to 0.530).

The procedure also isolates what fine-tuning buys. Recalibrated on its own evidence and scored on held-out drives of the search videos, the base checkpoint stays significantly below the fine-tuned model, by 0.117 macro-F1 (Table II); its collapse under the fine-tuned thresholds (Section V-B) overstates the gap.

A matched search budget shrinks the detector's recalibration gain but does not remove it. All choices are made on calibration data, VLM thresholds on the 100 videos with recorded VLM evidence and detector thresholds and counted tables on all 312, so the budgets favour the detector. Given the VLM's budget (one grid, the same 100 videos), the recalibrated pipeline still gains significantly over its hand-tuned version and is level with C3E (Table II).

c) No configuration dominates.: Calibrated, the world model and the detector pipeline tie on macro-F1 (Table II) and fail differently. C3E leads INSIDE IoU and entry MAE; the detector pipeline leads precision and nuisance, with less than a third of C3E's false alarms (Table I). Section VII combines them.

VII. Combining the Detector and the World Model

The joint estimator ties the recalibrated pipeline, trading precision for recall: it finds more INSIDE events but raises more false alarms (Tables I and II). It significantly beats the published pipeline (+0.076 macro-F1) and C3E alone (+0.031).

a) Sensitivity.: Of the joint estimator's two settings, only τ matters: it trades accuracy for nuisance ($\tau = 0.25$ s gives 0.538 macro-F1 at 44.7 false alarms per hour, $\tau = 1$ s 0.504 at 22.0), while coarser or finer detector buckets move macro-F1 by less than 0.01.

b) Concurrent execution.: The replay is faithful under real GPU contention. Running both models truly concurrently on the Orin, in wall-clock time from a decoded camera stream on all validation videos, slows the detector cycle from 24 to 53 ms, close to the 44 ms of the recorded runs behind every result, and the world model answers about every 1.5 s in both. Live, the joint estimator scores 0.548 against 0.546 in replay ($+0.002$, $[-0.011, +0.020]$) with no significant change in false alarms, and the detector-only filter 0.522 against 0.524 (Table II).

c) Where the gain comes from.: The detector carries the filter: without it macro-F1 drops by 0.086 and EXITING almost vanishes (IoU 0.017), while removing the world model costs only 0.022. The detector score alone in the counted filter is the most efficient option here: without CLIP or a threshold search, it ties the recalibrated detector+CLIP pipeline and raises significantly fewer false alarms than a state machine recalibrated on the same score. The world model's edge shows on the ten validation onsets preceded by a transcribed work-zone sign: the joint estimator and C3E warned $\geq$2 s ahead on all ten, the recalibrated pipeline on eight.

d) A learned alternative.: A causal three-stage TCN [24] on the same evidence ties the best macro-F1 (0.550) at its native false-alarm rate, 71.9 per hour, nearly three times the joint estimator's.

VIII. Do the Conclusions Hold?

a) Unseen drives.: The conclusions hold on drives a system was never fitted on. Splitting the 25 drives into five folds of about 104 videos, we refit the detector state machines (full three-grid search) and the counted filters on four folds and replay them on the fifth (Table II). Recalibration still raises the detector+CLIP pipeline and the detector-score state machine significantly, by 0.054 and 0.089 macro-F1, and both counted filters tie the recalibrated pipeline.

The model finding holds on unseen drives too. With per-channel evidence now recorded for all 312 calibration videos, we search each VLM's constants on four of the same drive folds and replay them on the fifth (60–68 test videos). For C3E, every fold selects this paper's constants, except the SIGN switch in one fold; on held-out drives they lift it significantly over the 2B's constants, from 0.268 to 0.485 macro-F1, at 12.9 more false alarms per hour. The same search moves the fine-tuned 2B only slightly over its own constants ($+0.025$, from 0.464 to 0.490): constants matter most when inherited from another model (Table II).

b) Matched nuisance.: The calibrated systems also lead when all systems run at the same false-alarm rate. Their native rates differ widely (Table I), so we apply the persistence filter of Section IV to all, until false alarms fall to 20 per hour. On unseen drives, the recalibrated pipeline, the joint estimator and the counted detector filter then keep 0.52–0.54 macro-F1, against 0.42 for the hand-tuned pipeline (F1@20 on validation: Table I). Matching nuisance also reorders anticipation (Lead@20 in Table I). The VLM systems and the recalibrated pipeline need a longer persistence, which delays their warnings, while the counted detector filter needs the shortest one and most often warns early.

IX. Moving to a New Platform

The Thor runs every model faster than the Orin, while answers agree on 96–100% of identical frames: the change lies in timing, and replaying the hand-tuned pipeline's Thor recording reproduces its live run (Table II). Cycles shrink from 24 to 16 ms for the detector (35 to 21 ms median on the detector+CLIP recordings), 238 to 184 ms for the 2B's patrol cycle, 718 to 240 ms for C3E's and 44 to 12 ms for the joint loop.

a) Carried-over cycle constants lose accuracy.: Constants counted in cycles span less time on a faster device, and the loss tracks how much each cycle shrank. C3E's patrol cycle shrinks threefold and it loses 0.042 macro-F1, the detector's shrinks by a third (-0.031), and the 2B's least (a tie). False alarms rise for all three.

b) The recalibrated pipeline transfers, its alarm rate does not.: The Orin-recalibrated pipeline keeps its macro-F1 on the Thor (a tie), and re-searching the same grids on a Thor recording of the calibration videos selects exactly the same constants. It still beats the hand-tuned pipeline significantly there, by 0.099, but its false alarms rise significantly, by 15.2 per hour, a change a macro-F1 search cannot see.

c) Per-second filters transfer.: The joint estimator's loop shrinks most, almost fourfold, yet it keeps its macro-F1, while its false alarms rise by a similar fraction as the 2B's ($+37\%$ vs. $+38\%$). The detector-only filter, scored on the detector+CLIP recordings of both devices (on the Orin 0.524 and 25.3 false alarms per hour; Table I's 22.5 comes from the joint run's detector stream), keeps its macro-F1 (-0.002, $[-0.005, 0.000]$) with significantly fewer false alarms, and on the Thor it ties the recalibrated pipeline with 14.1 fewer per hour, a significant gap.

X. Failure Analysis: the Ego-Relevance Gap

Most residual false positives we inspected are correct descriptions of objects irrelevant to the vehicle's path: a parked work vehicle 60 m away, sidewalk work behind barriers, detour signs for another street. The qualifier filter catches only what the model verbalizes, and binary or distance prompts do not recover the judgment.

a) Response bias.: Recording three checkpoints on the same 100 calibration videos at 1 Hz shows that fine-tuning made the gate say "yes" more often while tracking the state worse; the fine-tuned model's advantage comes from its description. The base 2B's gate says "yes" on 3.8% of OUTSIDE seconds and separates INSIDE from OUTSIDE best (balanced accuracy 0.84); fine-tuning raises that rate to 42.4% and lowers balanced accuracy to 0.75. Since ROADWork's OUTSIDE includes visible but irrelevant work, such a "yes" is often correct perception without ego-relevance. The fine-tuned description, in contrast, corroborates on 10.7% of OUTSIDE seconds against 31.3% for the base model.

C3E, never fine-tuned (gate "yes" on 29.4% of OUTSIDE seconds, balanced accuracy 0.82), transcribes signs correctly yet labels most verified INSIDE frames "off-road/sidewalk" in ego-lane probes. Perceiving objects and judging whether they claim the ego path are distinct capabilities, and no prompt we tried bridged them in either model. The joint estimator tracks EXITING only through the detector, whose score decays as objects recede (Section VII). ROADWork's 129K path annotations [5] could supply trajectory-grounded ego-relevance labels.

XI. Limitations

Two limits bound our claims. *(i)* All footage is ROADWork daytime driving, and the 2B's fine-tuning images may share drives with the benchmark (Section IV). *(ii)* Replay is open-loop: we show computational feasibility, not safety readiness.

XII. Conclusion

This study yields three insights. First, temporal constants belong to a model, not to a task: inherited across models they move INSIDE IoU by 0.567, an effect that holds on unseen drives. Second, the baseline must be calibrated like the proposed system: doing so lifts it from 0.470 to 0.551 and removes the VLM-based systems' lead, and a counted filter on its score alone is not significantly different from it on unseen drives. Third, temporal constants also belong to a platform: on the faster Thor, carried-over constants cost up to 0.04 macro-F1. The recalibrated pipeline, whose optimum is the same on both devices, and the per-second filters keep their macro-F1 there, but false alarms change unseen by a macro-F1 search.

Teams should recalibrate every system on the target platform, or state its constants in seconds, compare at matched nuisance, and recheck false alarms after a platform change. The VLM-based systems' edge is earlier warning on signled onsets, at a modest power cost (+5.5 W for C3E); at matched nuisance the per-second detector-only counted filter warns early most often and is the efficient alternative on both platforms. The VLMs' remaining gap is ego-relevance, which trajectory-grounded supervision may close; night and adverse-weather evaluation is the next step. Upon acceptance we release the replay harness, prompts, budgets, lexicons, metrics, runtime patches and baseline port.

References

[1] National Highway Traffic Safety Administration, "Work zone crashes and fatalities, 2023 (FARS and CRSS)," via National Work Zone Safety Information Clearinghouse, 2023.

[2] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[3] A. Martinez-Sanchez *et al.*, "ZONEWISE: Zone worksite inference with speed estimation," in *Proc. Int. Tech. Conf. Enhanced Safety of Vehicles (ESV)*, 2026.

[4] ——, "Vision-language work zone intelligence for safety-critical speed regulation of mixed-autonomy vehicles in dynamic environments," *arXiv preprint arXiv:2606.08860*, 2026.

[5] A. Ghosh *et al.*, "ROADWork: A dataset and benchmark for learning to recognize, observe, analyze and drive through work zones," in *Proc. ICCV*, 2025, pp. 6132–6142.

[6] J. Lin *et al.*, "AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration," in *Proc. MLSys*, 2024.

[7] NVIDIA, "TensorRT Edge-LLM: High-performance LLM and VLM inference runtime for embedded platforms," https://github.com/NVIDIA/TensorRT-Edge-LLM, 2026.

[8] W. Shi and R. R. Rajkumar, "Work zone detection for autonomous vehicles," in *Proc. IEEE ITSC*, 2021, pp. 1585–1591.

[9] F. Abodo *et al.*, "Detecting work zones in SHRP 2 NDS videos using deep learning-based computer vision," in *Proc. IEEE ICMLA*, 2018, pp. 679–686.

[10] J.-J. Hwang *et al.*, "EMMA: End-to-end multimodal model for autonomous driving," *arXiv preprint arXiv:2410.23262*, 2024.

[11] X. Tian *et al.*, "DriveVLM: The convergence of autonomous driving and large vision-language models," in *Proc. CoRL*, 2024.

[12] A.-M. Marcu *et al.*, "LingoQA: Visual question answering for autonomous driving," in *Proc. ECCV*, 2024.

[13] NVIDIA, "Cosmos-Reason1: From physical common sense to embodied reasoning," *arXiv preprint arXiv:2503.15558*, 2025.

[14] ——, "Cosmos 3: Omnimodal world models for physical AI," *arXiv preprint arXiv:2606.02800*, 2026.

[15] ——, "Alpamayo-R1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail," *arXiv preprint arXiv:2511.00088*, 2025.

[16] P. Viola and M. Jones, "Rapid object detection using a boosted cascade of simple features," in *Proc. CVPR*, 2001.

[17] L. Chen, M. Zaharia, and J. Zou, "FrugalGPT: How to use large language models while reducing cost and improving performance," *arXiv preprint arXiv:2305.05176*, 2023.

[18] G. Melis, C. Dyer, and P. Blunsom, "On the state of the art of evaluation in neural language models," in *Proc. ICLR*, 2018.

[19] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.

[20] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. MIT Press, 2005.

[21] Qwen Team, "Qwen3-VL: A versatile vision-language model series," https://github.com/QwenLM/Qwen3-VL, 2025.

[22] NVIDIA, "Cosmos-Reason2-2B model card," https://huggingface.co/nvidia/Cosmos-Reason2-2B, 2025.

[23] Federal Highway Administration, "Manual on uniform traffic control devices for streets and highways (MUTCD), part 6: Temporary traffic control," U.S. DOT, 2009.

[24] Y. A. Farha and J. Gall, "MS-TCN: Multi-stage temporal convolutional network for action segmentation," in *Proc. CVPR*, 2019, pp. 3575–3584.